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References:

- "Industrial Guide to Ergonomics Engineering", Dr. Osama Mohammed Elmardi Suleiman Khayal
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Faculty HOD Dean of Academics 13/1/26

Lecture-1

HUMAN PERFORMANCE AND LIMITATIONS

HUMAN PERFORMANCE AS PART OF THE MAINTENANCE ENGINEERING SYSTEM

- **Mechanical Components:**
  - Have defined performance limits (e.g., rivets withstand force but can fail under excessive stress).
  - Can suffer **catastrophic failures** if overstressed.
- **Human Capabilities and Limitations:**
  - **Physical limitations:** Fatigue, exposure to cold, injuries (e.g., broken bones).
  - **Mental limitations:** Errors, limited perception, poor judgment due to lack of skills/knowledge.
  - **Social & emotional factors** also impact performance.
- **Comparison of Human Capabilities and Limitations with Mechanical Components:**
  - Human performance degrades under stress, similar to mechanical failure.
  - Unlike machines, humans are influenced by emotional and social conditions.
- **Impact on Aircraft Safety:**
  - Aircraft engineers play a **critical role** in the maintenance system.
  - Understanding human limitations helps improve **effectiveness and safety**.

Human Factors leading to accidents/incidents

Investigation revealed that, all of the incidents were characterized by the following:

- There were staff shortages
- Time pressures existed
- All the errors occurred at night
- Shift or task handovers were involved
- They all involved supervisors doing long hands-on tasks
- There was an element of a "can-do " attitude
- Interruptions occurred

- There was some failure to use approved data or company procedures
- Manuals were confusing
- There was inadequate pre-planning, equipment or spares

Vision

Vision plays a crucial role in aviation, as it is essential for pilots, air traffic controllers, and maintenance engineers to ensure safe and efficient aircraft operations. In aviation systems, vision is used for navigation, situational awareness, and decision-making. However, various factors can affect visual performance, leading to potential safety risks

The eye functions like a camera, with an iris (aperture), lens, and retina (light-sensitive surface). Light enters through the cornea, passes through the iris and lens, and reaches the retina. Rods and cones convert light into electrical impulses, sent via the optic nerve to the brain for interpretation.

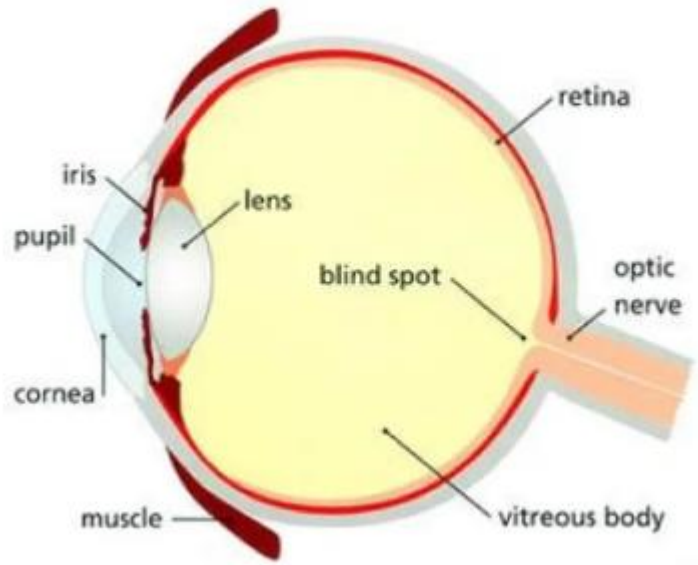


Fig-1 Human eye

Key Parts of the Eye and Their Functions

- **Cornea:** Transparent front part of the eye, responsible for 70-80% of total focusing by bending light.

- **Iris & Pupil:** The iris controls light entry by adjusting pupil size (5:1 ratio for light regulation).
- **Lens:** Adjusts shape (accommodation) for final focusing; thickens for near objects, flattens for distant ones. Affected by fatigue and aging.

  When a person is tired accommodation is reduced, resulting in less sharp vision (sharpness of vision is known as visual acuity)

- **Retina:** Located at the back of the eye, contains nerve cells linked to the optic nerve.
  1. **Light-Sensitive Cells:**
     - **Cones** (central fovea) – for sharp, color vision.
     - **Rods** (periphery) – for low-light and peripheral vision.

  2. **Blind Spot:** Where the optic nerve exits; unnoticed due to binocular vision and rapid eye movements (saccades).
  3. **Engineering Relevance:** Visual scanning prevents blind spot issues in tasks like crack detection.

An individual with acuity of 20/20 vision should be able to see at 20 feet that which the so-called 'normal ' person is capable of seeing at this range. It may be expressed in meters as 6/6 vision. The figures 20/40 mean that the observer can read at 20 feet that a 'normal ' person can read at 40 feet.

Various factors can affect and limit the visual acuity of the eye. These include:

1. Physical factors such as:
   - ♣ age.
   - ♣ physical imperfections in one or both eyes (short sightedness, long sightedness).
2. The influence of injested foreign substances such as:
   - ♣ drags
   - ♣ medication,
   - ♣ alcohol,
   - ♣ cigarettes.
3. Environmental factors such as:
   - ♣ amount of light available;
   - ♣ clarity of the air (e.g. dust, mist, rain, etc.).
4. Factors associated with object being viewed such as:
   - ♣ size and contours of the object,
   - ♣ contrast of the object with its surroundings,
   - ♣ relative motion of the object,
   - ♣ distance of the object from the viewer,

♣ the angle of the object from the viewer.

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
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Lecture-2

Myopia And Hypermetropia

Long sight: known as Hypermetropia is caused by a shorter than normal eyeball which means that the image is formed behind the retina. If the cornea and the lens cannot use their combined focusing ability to compensate for this, blurred vision will result when looking at close objects. A convex lens will overcome long sightedness by bending light inwards before it reaches the cornea.

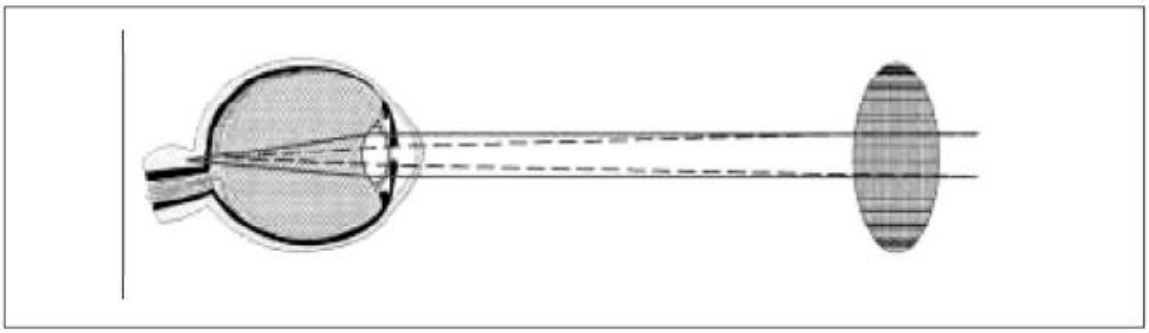


Fig-2 A convex lens will overcome long sightedness by bending light inwards before it reaches the cornea.

Short sight, known as Myopia is where the eyeball is longer than normal, causing the image to be formed in front of the retina. If the accommodation of the lens cannot counteract this then distant objects are blurred. A concave lens will overcome short sightedness by bending light outwards before it overshoots the cornea.

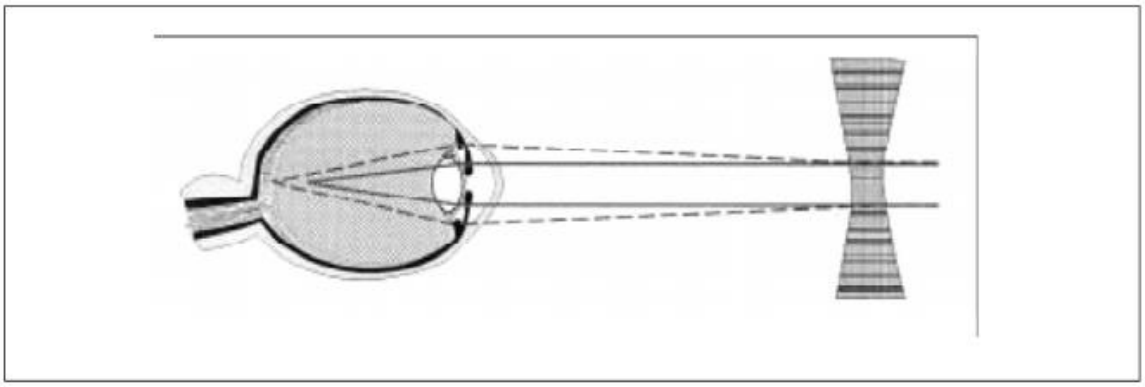


Fig-3 A concave lens will overcome short sightedness by bending light outwards before it reaches the cornea.

Other visual problems include:

- cataracts - clouding of the lens usually associated with ageing;
- astigmatism - a misshapen cornea causing objects to appear irregularly shaped;

- glaucoma - a build up in pressure of the fluid within the eye which can cause damage to the optic nerve and even blindness;
- migraine - severe headaches that can cause visual disturbances.

Colour Vision

While colour vision does not directly impact visual acuity, it is crucial for aircraft maintenance engineers in tasks such as component recognition, wire differentiation, diagnostic tool usage, and identifying airfield warning lights.

Colour-defective vision, often hereditary, affects about 8% of men and 0.5% of women, with red-green deficiency being the most common. Some may also struggle with blue-yellow distinction. The severity varies, and low light can worsen the condition. With age, colour perception declines due to lens yellowing, mainly affecting blue-yellow discrimination.

Vision and the Aircraft Maintenance Engineer

The UK CAA (Civil Aviation Authority) advises that aircraft engineers require good near and distance vision, especially for inspecting structures and handling small components. Corrective eyewear is acceptable and must be used as prescribed, with regular checks for effectiveness. Colour discrimination is crucial for tasks like wiring and airfield driving. Engineers rely on vision to inspect aircraft components for damage, corrosion, or wear.

Airlines and airports may impose additional eyesight standards for insurance or operational reasons. Ultimately, individuals must recognize any vision impairment and assess its impact on their work.

Hearing

Hearing is a crucial sensory ability in aviation, especially for pilots, air traffic controllers, and aircraft maintenance engineers. It enables effective communication, situational awareness, and safety

Basic Function of the Ear

The ear serves two main functions: detecting sound and maintaining balance. For maintenance engineers, hearing is the primary concern.

Structure of the Ear

- **Outer Ear:** Directs sound waves to the eardrum, causing vibrations**.**
- **Middle Ear**: Contains three ossicle bones that transmit vibrations to the inner ear. It also has muscles that reduce noise above 80 dB for up to 15 minutes. The eustachian tube regulates air pressure but can allow mucus buildup, affecting hearing.
- **Inner Ear:** Filled with fluid, it contains the cochlea, where hair-like cells detect vibrations and send neural impulses to the brain**.**

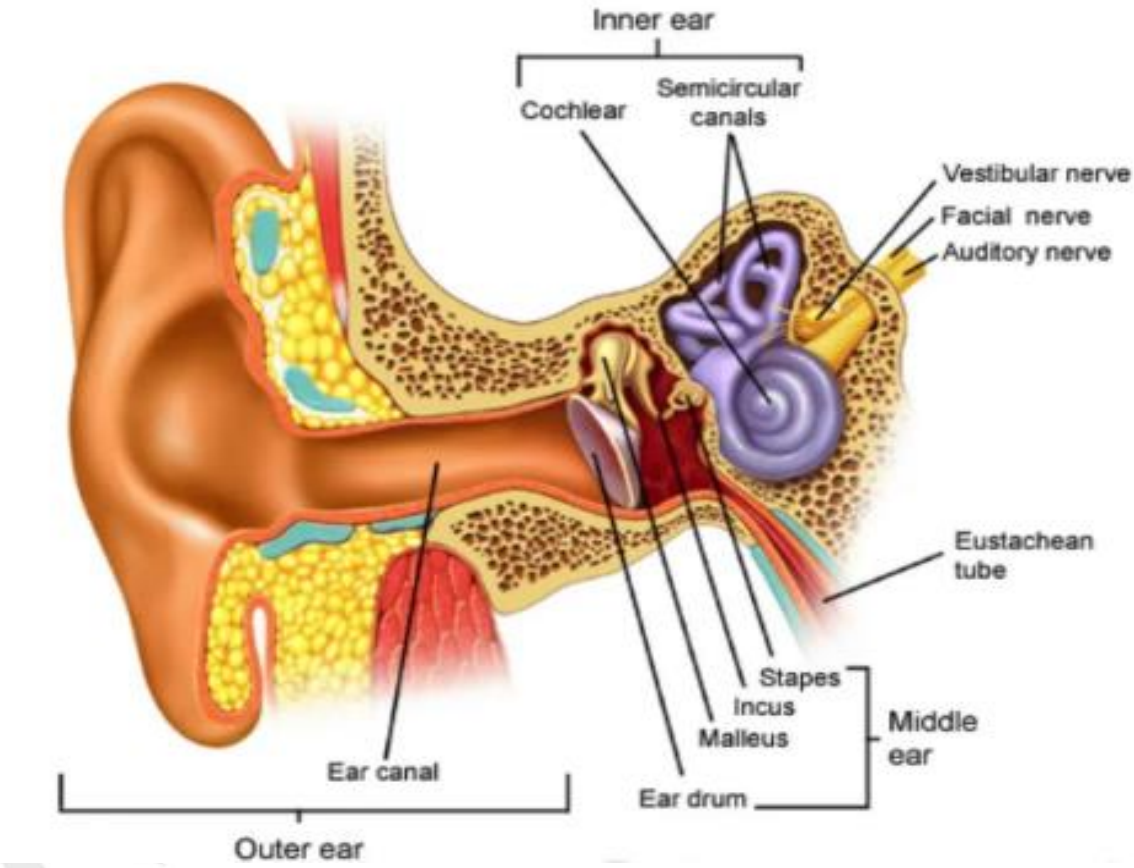


Fig-4 The human ear

Performance capabilities include:

1. **Sound Detection:**
   - Recognizing alarms, radio communications, and mechanical noises.
   - Identifying abnormal sounds in aircraft components, such as engine malfunctions or warning signals.
2. **Direction and Distance Perception:**

- Localizing sound sources helps pilots and engineers detect issues.
- Differentiating between close and distant noises is essential for safety.

3. **Speech Recognition in Noisy Environments:**
   - Aviation professionals must understand radio transmissions despite background noise.
   - Clear communication prevents misunderstandings and errors.

Limitations of Human Hearing in Aviation

Despite its importance, hearing has several limitations that can impact safety and efficiency in aviation:

1. Hearing Loss and Fatigue:
2. Masking Effect:
3. Limited Frequency Range:
4. Aging and Health Factors:

Mitigation Strategies

- Use of noise-cancelling headsets and hearing protection.
- Regular hearing tests and monitoring.
- Training to improve auditory discrimination in noisy environments.
- Enhancing communication protocols, such as using standardized phraseology in radio transmissions.

Reference

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Lecture-3

Impact of Noise on Performance

Noise can have various negative effects in the workplace. It can:

• be annoying (e.g. sudden sounds, constant loud sound, etc.);

• interfere with verbal communication between individuals in the workplace;

• cause accidents by masking warning signals or messages;

• be fatiguing and affect concentration, decision making, etc.;

• damage workers' hearing (either temporarily or permanently

Hearing Impairment & Protection

Noise-Induced Hearing Loss (NIHL)

Hearing loss can result from short or prolonged exposure to loud noise. Temporary loss occurs when hair cells in the cochlea take time to recover, but repeated exposure leads to permanent damage.

UK 'Noise at Work' Regulations (1989)

Employers must act at:

- 85 dB: Assess risks, inform employees, and provide ear protection.
- 90 dB: Reduce noise exposure and mark restricted zones.
- 140 dB: Causes pain; strict protection required.

A Time-Weighted Average (TWA) above 85 dB for 8 hours is hazardous. Exposure over 115 dB without protection is strongly discouraged.

Hearing Protection

- Earplugs reduce noise by up to 20 dB.
- Earmuffs reduce noise by up to 40 dB. Despite affecting communication, consistent use is essential. Reducing noise at the source is ideal but often impractical in aviation maintenance.

Presbycusis (Age-Related Hearing Loss)

Presbycusis is the gradual loss of hearing that occurs as people age. It's a common condition, affecting about one in three adults between the ages of 65 and 74, and nearly half of those older than 75.

This type of hearing loss typically affects both ears and progresses slowly, making it difficult for individuals to notice the change initially. Common symptoms include difficulty hearing high-pitched sounds, trouble understanding speech, especially in noisy environments, and the perception that sounds are muffled. Presbycusis is primarily caused by age-related changes in the inner ear, but factors such as genetics, long-term exposure to loud noises, and certain medical conditions can also contribute

Hearing and the Aircraft Maintenance Engineer

The UK CAA1 makes the following recommendations regarding hearing:

“The ability to hear an average conversational voice in a quiet room at a distance of 2 metres (6 feet) from the examiner is recommended as a routine test. Failure of this test would require an audiogram to be carried out to provide an objective assessment. If necessary, a hearing aid may be worn but consideration should be given to the practicalities of wearing the aid during routine tasks demanded of the individual.”

Aircraft maintenance engineers operate in environments with noise levels that can reach up to 140 decibels, particularly near jet engines and during the use of pneumatic tools. Exposure to noise levels above 85 decibels can cause permanent hearing damage over time.

Reference

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Lecture-4

Information processing

Information processing is the process of receiving information through the senses, analysing it and making it meaningful. Information processing can be represented as a model. This captures the main elements of the process, from receipt of information via the senses, to outputs such as decision making and actions

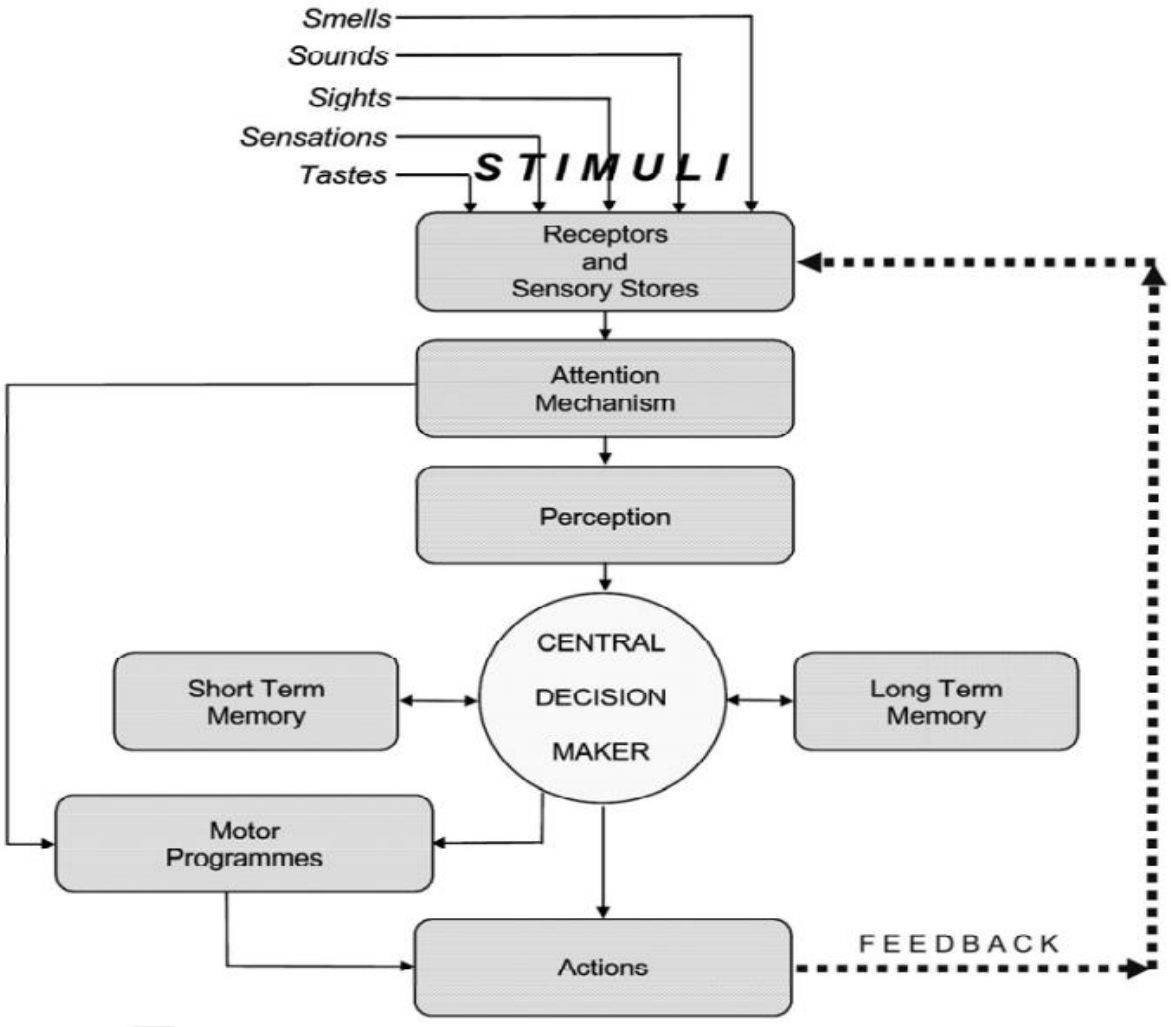


Fig-5 A functional model of human information processing

Explanation of the Diagram:

1. **Stimuli (External Inputs)**

   Information from the environment is received through different senses:

   - **Smells** (Olfactory)

- **Sounds** (Auditory)
- **Sights** (Visual)
- **Sensations** (Tactile)
- **Tastes** (Gustatory)

2. **Receptors and Sensory Stores**
    - These stimuli are first processed by **sensory receptors** (eyes, ears, skin, nose, tongue).
    - Sensory memory **briefly holds** this information before passing it further into the system.
3. **Attention Mechanism**
    - The brain filters and selects **important stimuli** for further processing.
    - Irrelevant or less significant stimuli may be ignored.
4. **Perception**
    - The selected information is **interpreted and understood** by the brain.
    - The brain organizes this raw sensory data into **meaningful patterns**.
5. **Central Decision Maker**
    - This is the **main cognitive processing unit** where:
        - Incoming information is analyzed.
        - It is compared with past experiences.
        - A decision is made based on available information.
6. **Memory Systems**
    - The **Short-Term Memory (Working Memory)** temporarily holds information for immediate use.
    - The **Long-Term Memory** stores past experiences, knowledge, and learned responses for future use.
7. **Motor Programmes**
    - Once a decision is made, motor programmes are activated.
    - These are pre-learned sequences of movements (e.g., typing, driving, or speaking).
8. **Actions**

- The decision results in a **physical or verbal action** (e.g., pressing a button, speaking a word, moving away from danger).

9. **Feedback Mechanism**
    - The system continuously monitors actions and their outcomes.
    - Feedback loops help improve **future responses** and **adjust actions if needed**.

Attention and Perception

Attention can be thought of as the concentration of mental effort on sensory or mental events. Although attention can move very quickly from one item to another, it can only deal with one item at a time. Attention types:

1.Selective attention

2.Divided attention

3.Focused attention

4.Sustained attention.

Selective attention occurs when a person is monitoring several sources of input, with greater attention being given to one or more sources which appear more important A classic example is the "cocktail party effect," where a person can engage in a conversation amidst background noise yet still notice if someone across the room mentions their name even though you were not aware of listening in to other people's conversations. Distraction is the negative side of selective attention.

Divided attention is common in most work situations, where people are required to do more than one thing at the same time. Usually, one task suffers at the expense of the other, more so if they are similar in nature. This type of situation is also sometimes referred to as time sharing.

Focused attention is merely the skill of focusing one's attention upon a single source and avoiding distraction.

Sustained attention as its name implies, refers to the ability to maintain attention and remain alert over long periods of time, often on one task. Most of the research has been carried out in connection with monitoring radar displays, but there is also associated research which has concentrated upon inspection tasks.

Perception can be defined as the process of assembling sensations into a useable mental representation of the world. Perception creates faces, melodies, works of art, illusions, etc. out of the raw material of sensation

Examples

• the image formed on the retina is inverted and two dimensional, yet we see the world the right way up and in three dimensions;

• if the head is turned, the eyes detect a constantly changing pattern of images, yet we perceive things around us to have a set location, rather than move chaotically

Decision Making

Decision making is the generation of alternative courses of action based on available information, knowledge, prior experience, expectation, context, goals, etc. and selecting one preferred option. It is also described as thinking, problem solving and judgement.

Memory

Memory can be considered to be the storage and retention of information, experiences and knowledge, as well as the ability to retrieve this information

Memory depends on three processes:

• registration - the input of information into memory;

• storage - the retention of information;

• retrieval - the recovery of stored information.

It is possible to distinguish between three forms of memory:

a. ultra short-term memory (or sensory storage);

b. short term memory (often referred to as working memory)

c. long term memory

ultra short-term memory - It has a duration of up to 2 seconds (depending on the sense) and is used as a buffer, giving us time to attend to sensory input

short term memory is a cognitive system responsible for temporarily holding and processing information needed for complex tasks such as learning, reasoning, and comprehension. It has a limited capacity, typically holding about 7±2 items, and retains information for a brief duration, usually around 20 to 30 seconds, unless actively rehearsed. This system allows individuals to manipulate and utilize information in real-time, facilitating daily cognitive functions,

The capacity of long-term memory appears to be unlimited. It is used to store information that is not currently being used, including:

• knowledge of the physical world and objects within it and how these behave;

• personal experiences;

• beliefs about people, social norms, values, etc.;

• motor programmes, problem solving skills and plans for achieving various activities;

• abilities, such as language comprehension.

Information in long-term memory can be divided into two types:

(i) Semantic memory- refers to our store of general, factual knowledge about the world, such as concepts, rules, one's own language, etc. It is information that is not tied to where and when the knowledge was originally acquired

(ii) Episodic memory -refers to memory of specific events, such as our past experiences (including people, events and objects). We can usually place these things within a certain context. It is believed that episodic memory is heavily influenced by a person's expectations of what should have happened, thus two people's recollection of the same event can differ

Reference

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Lecture-5

Physical Access and Claustrophobia

1.Claustrophobia

There are many circumstances where people may experience various levels of physical or psychological discomfort when in an enclosed or small space, which is generally considered to be quite normal. When this discomfort becomes extreme, it is known as claustrophobia.

2.Fear of Heights

Working at significant heights during aircraft maintenance, such as performing 'crown' inspections atop the fuselage, can pose challenges for engineers, especially those with a fear of heights. Key considerations include:

1.**Individual Comfort Levels**: While some engineers are comfortable working at heights, others may experience discomfort or anxiety, which can distract from the task at hand.

2.**Use of Safety Equipment**: Employing appropriate safety measures, such as harnesses and safety ropes, is crucial. These tools not only enhance safety but also provide reassurance, helping engineers focus better on their tasks.

3.**Access Equipment Guidance**: The Federal Aviation Administration (FAA) offers practical guidance on the use of access equipment when working at height. Engineers should familiarize themselves with these resources to ensure safe practices.

4.**Managing Severe Anxiety**: If an engineer experiences severe anxiety or panic symptoms when working at heights, it's advisable to avoid such situations for safety reasons. Support from team members can be beneficial in these cases.

3.environmental tolerances

In aviation, environmental tolerances refer to the ability of aircraft systems and components to withstand various environmental conditions encountered during operation. The RTCA DO-160 standard outlines specific test procedures to ensure equipment reliability under factors such as temperature variations, humidity, shock, vibration, and electromagnetic interference. Adhering to these standards is crucial for maintaining safety and performance in diverse operating environments

4. Humidity and pressure

Humidity: Elevated moisture levels in the air decrease air density, leading to reduced lift and engine efficiency. This necessitates longer takeoff distances and can impair climb rates.

Pressure: Variations in atmospheric pressure alter air density, directly affecting lift and engine thrust. Lower pressure results in decreased air density, diminishing aircraft performance

Reference

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